

## NORMANTON, Australia

WMO: 942660

Lat: 17.6869S

Lon: 141.0733E

Elev: 62

StdP: 14.66

Time Zone: 10.00 (E10)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 56.4       | 58.8 | 19.2                          | 14.5 | 75.6 | 23.0 | 17.4 | 74.2 | 21.1                  | 66.8 | 19.6 | 68.0 | 9.9                   | 170  | 0.397 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 11            | 17.6                   | 99.5            | 70.8 | 97.7 | 72.5 | 96.2 | 73.4 | 82.5                | 88.3 | 81.5 | 87.8 | 80.8 | 87.3 | 9.5                  | 310  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.2                            | 162.6 | 84.8 | 80.1 | 156.4 | 84.3 | 79.2 | 151.8 | 83.9 | 46.5          | 88.7 | 45.4 | 87.5 | 44.6 | 87.4 | 90.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 18.8          | 17.0 | 15.6 | 52.9                       | 104.3 | 2.4                | 2.1 | 51.1                                               | 105.8 | 49.7       | 107.1 | 48.3       | 108.2 | 46.6       | 109.8 |    |    |
| (5)               |      |      | 42.0                       | 83.9  | 1.5                | 2.2 | 41.0                                               | 85.4  | 40.1       | 86.7  | 39.3       | 87.9  | 38.3       | 89.5  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 81.3   | 84.7  | 84.6  | 84.3  | 82.7  | 77.8  | 73.5  | 73.1  | 75.3  | 80.9  | 85.4  | 86.9  | 86.8  |
|                                                                                  | DBStd       | 6.03   | 3.32  | 2.91  | 2.90  | 3.22  | 4.18  | 4.87  | 4.43  | 3.91  | 3.20  | 2.69  | 2.69  | 3.14  |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65       | 7      | 0     | 0     | 0     | 0     | 0     | 4     | 3     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50       | 11432  | 1075  | 969   | 1063  | 982   | 863   | 706   | 717   | 785   | 928   | 1096  | 1107  | 1141  |
|                                                                                  | CDD65       | 5964   | 610   | 549   | 598   | 532   | 398   | 260   | 255   | 320   | 478   | 631   | 657   | 676   |
|                                                                                  | CDH74       | 69164  | 7471  | 6690  | 7251  | 6144  | 3737  | 2106  | 2125  | 2959  | 5098  | 7900  | 8693  | 8990  |
|                                                                                  | CDH80       | 32843  | 3457  | 3042  | 3312  | 2854  | 1496  | 682   | 670   | 1167  | 2525  | 4181  | 4670  | 4787  |
| Wind                                                                             | WSAvg       | 7.7    | 6.8   | 6.2   | 6.2   | 6.7   | 7.7   | 8.4   | 8.3   | 8.4   | 8.5   | 8.6   | 8.3   | 7.8   |
| Precipitation                                                                    | PrecAvg     | 33.6   | 9.0   | 8.4   | 6.1   | 1.1   | 0.2   | 0.3   | 0.1   | 0.0   | 0.2   | 0.5   | 1.8   | 5.4   |
|                                                                                  | PrecMax     | 72.9   | 38.0  | 20.1  | 18.7  | 4.8   | 1.5   | 3.3   | 0.9   | 0.6   | 2.5   | 3.1   | 9.1   | 15.8  |
|                                                                                  | PrecMin     | 17.5   | 1.7   | 0.5   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.2   |
|                                                                                  | PrecStd     | 13.4   | 7.5   | 5.1   | 3.9   | 1.3   | 0.4   | 0.7   | 0.2   | 0.1   | 0.5   | 0.7   | 1.9   | 3.7   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 99.2  | 98.1  | 97.9  | 96.6  | 93.8  | 90.4  | 90.1  | 94.0  | 97.7  | 101.2 | 102.1 | 102.2 |
|                                                                                  |             | MCWB   | 73.2  | 77.1  | 73.5  | 72.1  | 70.6  | 68.6  | 68.9  | 68.4  | 66.9  | 68.1  | 66.4  | 71.0  |
|                                                                                  | 2%          | DB     | 96.2  | 95.5  | 95.2  | 94.5  | 91.3  | 88.1  | 87.7  | 90.8  | 95.3  | 98.2  | 99.0  | 98.7  |
|                                                                                  |             | MCWB   | 77.1  | 77.6  | 75.5  | 72.3  | 69.7  | 67.8  | 67.0  | 66.0  | 67.8  | 69.7  | 70.2  | 73.7  |
|                                                                                  | 5%          | DB     | 94.1  | 93.8  | 93.5  | 93.0  | 89.5  | 86.4  | 86.0  | 88.6  | 93.3  | 96.3  | 97.0  | 96.6  |
|                                                                                  |             | MCWB   | 78.3  | 77.9  | 75.7  | 71.8  | 69.0  | 67.0  | 65.3  | 65.0  | 67.6  | 70.3  | 72.5  | 75.5  |
|                                                                                  | 10%         | DB     | 92.2  | 92.0  | 91.8  | 91.5  | 87.7  | 84.4  | 84.2  | 86.4  | 91.3  | 94.6  | 95.2  | 94.8  |
|                                                                                  |             | MCWB   | 78.4  | 78.3  | 75.7  | 71.6  | 68.3  | 65.8  | 64.2  | 63.9  | 67.1  | 70.4  | 73.9  | 76.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 84.9  | 85.0  | 82.1  | 79.7  | 75.9  | 72.5  | 71.9  | 72.5  | 75.8  | 80.8  | 82.1  | 82.5  |
|                                                                                  |             | MCDB   | 88.0  | 90.1  | 87.2  | 86.9  | 85.8  | 83.9  | 85.2  | 86.5  | 88.6  | 90.8  | 88.6  | 88.7  |
|                                                                                  | 2%          | WB     | 82.7  | 82.1  | 80.9  | 78.3  | 74.1  | 70.7  | 70.0  | 70.5  | 74.4  | 78.7  | 80.5  | 81.4  |
|                                                                                  |             | MCDB   | 87.9  | 88.1  | 86.7  | 85.0  | 84.4  | 83.0  | 83.3  | 83.5  | 86.3  | 88.1  | 88.3  | 87.8  |
|                                                                                  | 5%          | WB     | 81.5  | 81.1  | 80.0  | 77.3  | 72.7  | 69.3  | 68.4  | 68.6  | 73.2  | 77.3  | 79.5  | 80.6  |
|                                                                                  |             | MCDB   | 87.7  | 87.3  | 86.0  | 84.4  | 83.5  | 81.6  | 81.2  | 82.3  | 84.3  | 86.4  | 87.5  | 87.3  |
|                                                                                  | 10%         | WB     | 80.6  | 80.4  | 79.2  | 76.3  | 71.0  | 67.9  | 66.6  | 67.0  | 72.0  | 76.2  | 78.5  | 79.8  |
|                                                                                  |             | MCDB   | 87.3  | 86.7  | 85.5  | 84.1  | 82.8  | 81.0  | 80.0  | 81.0  | 83.1  | 85.7  | 86.9  | 87.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.2  | 13.2  | 14.2  | 17.4  | 19.5  | 20.4  | 21.3  | 23.0  | 22.5  | 20.4  | 17.6  | 15.6  |
|                                                                                  |             | MCDBR  | 16.4  | 16.5  | 17.1  | 19.0  | 20.3  | 21.1  | 22.0  | 24.1  | 23.9  | 22.1  | 20.0  | 18.6  |
|                                                                                  |             | MCWBR  | 5.4   | 5.2   | 5.6   | 6.5   | 8.0   | 9.5   | 9.8   | 9.8   | 8.7   | 8.1   | 7.6   | 6.4   |
|                                                                                  | 5% WB       | MCDBR  | 13.6  | 13.7  | 14.5  | 14.6  | 17.1  | 19.4  | 20.1  | 22.4  | 20.6  | 18.1  | 16.2  | 14.8  |
|                                                                                  |             | MCWBR  | 5.3   | 5.1   | 5.3   | 5.5   | 7.1   | 9.0   | 9.0   | 9.1   | 8.0   | 6.3   | 6.2   | 5.3   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.413  | 0.409 | 0.387 | 0.363 | 0.344 | 0.330 | 0.329 | 0.344 | 0.391 | 0.419 | 0.444 | 0.438 |       |
|                                                                                  | taud        | 2.482  | 2.495 | 2.550 | 2.575 | 2.556 | 2.557 | 2.539 | 2.474 | 2.356 | 2.310 | 2.297 | 2.358 |       |
|                                                                                  | Ebn at Noon | 296    | 294   | 294   | 289   | 284   | 282   | 286   | 291   | 287   | 287   | 285   | 288   |       |
|                                                                                  | Edh at Noon | 37     | 36    | 33    | 30    | 29    | 28    | 29    | 33    | 40    | 43    | 45    | 42    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1869   | 1923  | 1923  | 1850  | 1603  | 1473  | 1576  | 1833  | 2120  | 2275  | 2233  | 2071  |       |
|                                                                                  | RadStd      | 195    | 189   | 175   | 131   | 80    | 98    | 54    | 61    | 78    | 116   | 130   | 185   |       |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | +1.82   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page